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Guiding VI selection for phenology monitoring: Differential sensitivity of vegetation indices to temporal dynamics in canopy leaf area and pigment

Jiaxin Jin [a,b,*], Xiaolong Cheng [b], Yulong Cai [c], Yuanwei Qin [b], Qiuan Zhu [b], Weifeng Wang [d], Peiqi Yang [e], Jianbo Qi [f], Feng Zhou [b], Guishan Yang [b], Jin Wu [g,h,*]

[a] State Key Laboratory of Water Disaster Prevention, Hohai University, Nanjing 210024, China
[b] Jiangsu Key Laboratory of Soil and Water Processes in Watershed, College of Geography and Remote Sensing, Hohai University, Nanjing 211100, China
[c] Songliao River Water Resources Commission of Ministry of Water Resources, Changchun 130021, China
[d] College of Biology and the Environment, Nanjing Forestry University, Nanjing 210037, China
[e] School of Geography, Nanjing Normal University, Nanjing, China
[f] Faculty of Geographical Science, Beijing Normal University, Beijing, China
[g] School of Biological Sciences and Institute for Climate and Carbon Neutrality, The University of Hong Kong, Hong Kong 999007, China
[h] State Key Laboratory of Agrobiotechnology (CUHK), Hong Kong 999077, China

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ABSTRACT

Phenology, the study of recurring plant life-cycle events, is key to understand how terrestrial ecosystems respond to climate change. Remote sensing vegetation indices (VIs) have transformed large-scale monitoring of land surface phenology (LSP) by capturing seasonal shifts in vegetation activity. However, interpreting VI-derived phenological metrics such as start of growing season (SOS), which indicates spring phenology, remains challenging. This is because canopy spectral signals simultaneously reflect both canopy structural changes (e.g., leaf area expansion) and leaf biochemical changes (e.g., leaf pigment dynamics). To disentangle these influences, we integrated field measurements of leaf area index (LAI) and leaf chlorophyll content (LCC) with PROSAIL model simulations. The model generated time-series data for 21 widely used VIs under controlled LAI and LCC scenarios. Using five phenological extraction methods, we extracted SOS dates and systematically assessed how each VI responded to canopy leaf area versus leaf pigmental changes. Our results reveal a distinct divergence in VI behavior: 14 VIs (including NDII, EVI, PRI, CCI, GCC, NIRv, CCII, MSAVI, NDPI, SAVI, EVI2, MCARI, NDVI, and PI) primarily reflected canopy leaf area development, with NDII showing the strongest LAI sensitivity. Conversely, four VIs (MTCI, CVI, GCVI, and CIre) tracked pigment-related phenology linked to LCC. The remaining three (SR, NDRE, GNDVI) exhibited mixed but weaker sensitivity to both leaf area expansion and pigmental onset. Field observations and satellite data consistently validated these patterns. Importantly, our findings show that a VI's sensitivity is determined by its spectral band selection and mathematical formulation. This underscores the need to match VI choice to specific research goals, whether focusing on leaf biochemical properties or canopy structural development, for accurate LSP interpretation. We recommend that future phenological studies should explicitly distinguish between leaf area and pigmental phenophases to improve the ecological relevance of remote-sensing-based phenology in the context of climate change.

1. Introduction

Phenology, the timing of recurring biological events like spring leaf emergence, offers critical insights into how vegetation responds to seasonal climate shifts (Richardson et al., 2013), forming a critical link between climate dynamics and ecosystem processes (Buttò et al., 2023; Gu et al., 2022). Spring phenology, spanning events from budburst to canopy development, is particularly important, as it marks the start of growing season (SOS). Changes in SOS can profoundly influence ecosystem functioning, altering seasonal carbon uptake (Keenan et al., 2014), water use efficiency (Jin et al., 2017), and competitive interactions among plant species (Li et al., 2023). Accurate monitoring and biologically meaningful interpretation of spring phenology are therefore essential for predicting how terrestrial ecosystems respond to

* Corresponding author at: School of Biological Sciences and Institute for Climate and Carbon Neutrality, The University of Hong Kong, Hong Kong 999007, China.
E-mail addresses: jiaxinking@hhu.edu.cn (J. Jin), jinwu@hku.hk (J. Wu).

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environmental change (Caparros-Santiago et al., 2021; Piao et al., 2019).

Traditional ground-based monitoring methods are valuable in providing accurate phenological records, but are labor-intensive and limited in spatial and temporal coverage and consistency. By contrast, satellite remote sensing has transformed phenology research by enabling continuous, large-scale monitoring of land surface phenology (LSP) through its consistent spectral measurements, long-term time-series, and global coverage (Ma et al., 2022). Vegetation indices (VIs), calculated from spectral reflectance data, in particular, have demonstrated effectiveness in tracking seasonal changes in vegetation "greenness", and have thus been widely used to identify SOS and other key transitions across vast regions (Zeng et al., 2020). These VI-based phenophase metrics are now indispensable for studying climate-driven shifts in growing seasons globally and assessing associated vegetation-mediated climate feedbacks (Dronova and Taddeo, 2022; Fu et al., 2015; Lian et al., 2020;).

Despite these advances, a persistent challenge persists: different VIs often yield inconsistent SOS estimates. For example, He et al. (2015) found that SOS estimated from the Advanced Very High Resolution Radiometer (AVHRR) Global Inventory Modeling and Mapping Studies (GIMMS) NDVI dataset occurred approximately 19 days earlier than that estimated from the MERIS Terrestrial Chlorophyll Index (MTCI) in China. Similarly, studies in northern deciduous broadleaf forests have revealed systematic discrepancies, with SOS estimates based on structural VIs occurring earlier than that derived from a scaled chlorophyll/carotenoid index (sCCI) (Yin et al., 2020). While data processing choices can contribute to these differences, growing evidence (Guo et al., 2024; Yang et al., 2022; Yin et al., 2022) indicates that the underlying causes lies in the fundamental design of each VI—specifically, whether it responds more strongly to changes in canopy structure or leaf properties.

This points to a core mechanistic uncertainty: satellite-detected canopy reflectance, which is the basis for all VIs, simultaneously captures two distinct biological processes (Chen et al., 2022; Liu et al., 2023; Wu et al., 2018; Yang et al., 2014): Canopy structural development, e.g., measured by Leaf Area Index (LAI), influences light scattering within the canopy; meanwhile, biochemical changes such as shifts in Leaf Chlorophyll Content (LCC) affect the light absorption rate. Critically, these processes often follow different temporal trajectories in spring (Pantin et al., 2012). Field studies confirm that leaf expansion (LAI increase) typically precedes full chlorophyll maturation (LCC accumulation) in deciduous species (Croft et al., 2014, 2015). As illustrated in Fig. 1, similar reflectance signatures—for example, identical NDVI values—can result from either increased LAI or elevated LCC.

Furthermore, the sensitivity of a given VI to structural versus pigmental changes is determined by its spectral band selection and mathematical formulation. As such, a VI designed to detect chlorophyll will characterize SOS through the lens of leaf biochemistry, while a LAI-sensitive VI captures canopy leaf area development. Consequently, different indices may report different SOS dates because they capture different biological phenomena. This ambiguity could be compounded by the choice of phenology extraction method (e.g., threshold- vs. inflection point-based approaches), which also influences the biological meaning of SOS (Shang et al., 2017). Such uncertainties can propagate into climate impact studies, carbon cycle assessments, and ecosystem models that rely on SOS as a proxy for vegetation activity.

To address these issues, we systematically evaluated whether 21 common VIs track leaf area (LAI-driven) or pigmental (LCC-driven)

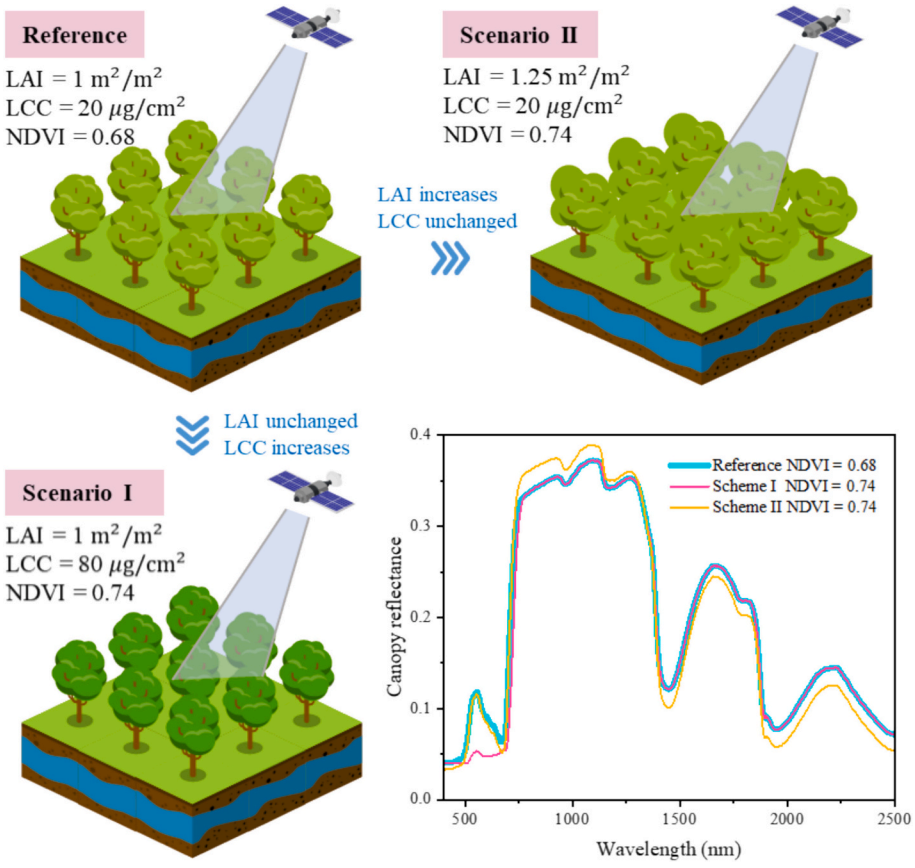


Fig. 1. A conceptual illustration of a canopy-scale greening of forest detected by satellite-based NDVI in two scenarios: an increasing LCC (= 80 μg/cm^2) with a constant LAI (= 1 m^2/m^2) (Scenario I), and a rising LAI (= 1.25 m^2/m^2) with a constant LCC (= 20 μg/cm^2) (Scenario II), comparing to the original LAI (= 1 m^2/m^2) and LCC (= 20 $\mu g/cm^2$) (Reference). The spectral reflectances in the different scenarios were simulated by the PROSAIL model. The extra parameters of the model are shown in Table 1.

spring phenology in Northern Hemisphere deciduous broadleaf forests (DBF). We integrated field-measured LAI and LCC data with the PROSAIL model to simulate canopy reflectance under controlled conditions. Two targeted experiments were conducted: 1) leaf area sensitivity test: LCC was held constant while LAI development timelines were shifted forward or backward by 1–10 days; 2) pigmental sensitivity test: LAI was held constant while LCC development timelines underwent equivalent shifts. For each scenario, we generated time series for all 21 VIs and extracted SOS dates using a standardized 20% threshold method (Jonsson and Eklundh, 2004). Additional sensitivity tests using alternative thresholds and other spring phenology extraction methods are presented in supplementary materials, with results consistent with the main analysis. By quantifying SOS shifts in response to LAI versus LCC changes, we determined: 1) the magnitude of SOS sensitivity to each driver; and 2) the dominant driver (LAI or LCC) for each VI. This approach isolates the influence of VI design on phenological interpretation, enabling researchers to select indices that align with specific objectives—whether monitoring leaf area expansion or biochemical traits. By linking VI design to biological interpretability, our study enhances the precision and ecological relevance of satellite-based phenology products, strengthening their role in tracking ecosystem responses to environmental change.

2. Materials and methods

2.1. Field observations

In this study, LAI and LCC were employed as proxies for canopy leaf area development and pigmental trait, respectively. These biophysical parameters were integrated into the PROSAIL model to simulate canopy reflectance spectra, from which multiple VIs were subsequently derived. The temporal trajectories of VIs were then analyzed to extract canopy-level SOS.

For numerical simulations, in-situ LAI and LCC datasets were compiled from three DBF sites across the Northern Hemisphere: 1) Harvard Forest, USA (42.53°N, 72.17°W; Yang et al., 2017), 2) De Inslag, Belgium (51.30°N, 4.51°E; Gond et al., 1999), and 3) New Forest National Park, UK (50.84°N, 1.57°W; Brown et al., 2019) (Fig. 2a). Field measurements spanned distinct growing seasons (2014, 1997, and 2016 respectively), ensuring biogeographical representativeness of mid-latitude DBF ecosystems. At Harvard Forest (dominated by *Quercus rubra*), LAI was quantified daily during spring/autumn and biweekly in summer using an LAI-2000 Plant Canopy Analyzer (LI-COR, Lincoln, NE). Parallel LCC determinations involved laboratory spectrophotometric analysis of leaf extracts (90% acetone with $MgCO_3$ stabilization) following identical temporal sampling intervals. The De Inslag site (*Quercus robur* stand) employed an 8-day LAI measurement protocol with the same instrumentation, while LCC was assessed through dimethylformamide (DMF) extraction at 15-day intervals during the February–December observational period. At New Forest National Park (*Fagus sylvatica* forest), LAI was derived from digital hemispherical photography using a Nikon Coolpix 4500 camera equipped with an FC-E8 fisheye lens (April–November 2016). LCC measurements were obtained non-destructively within several days of each LAI observation via a Konica Minolta SPAD-502 chlorophyll meter, which determines a relative value proportional to LCC. These values were then converted into absolute units using species-specific calibration functions (Brown et al., 2019).

A sigmoid-shaped logistic function (Eq. 1) was employed to parameterize the temporal dynamics of field-measured LAI and LCC across all three study sites (Fig. 2c-d):

$$y_t = \frac{c}{1 + e^{a+bt}} + d\# \tag{1}$$

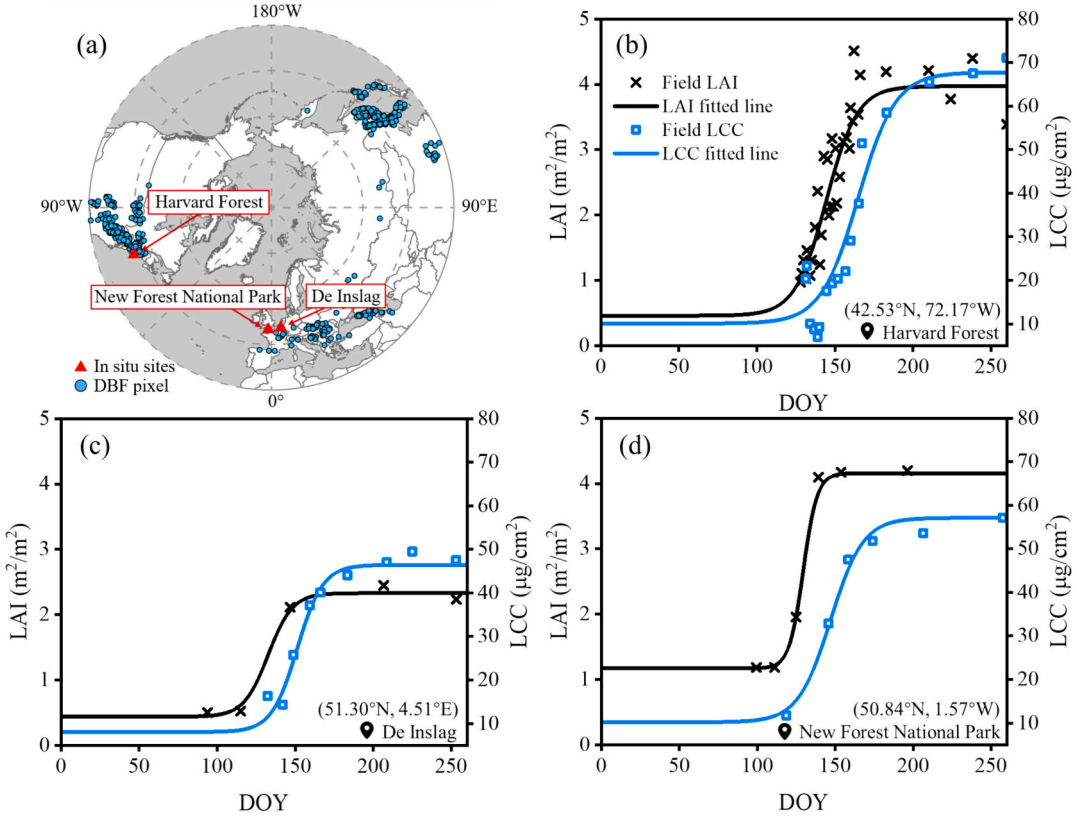


Fig. 2. Spatial distribution of (a) the three study sites and DBF samples obtained from MODIS land cover maps, and (b-d) the field observations of LAI and LCC (marked as black crosses and blue squares, respectively) with the corresponding fitted lines based on the logistic function. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where t represents day of year (DOY), and y_t corresponds to the LAI or LCC time series. Parameter a governs the phenological onset timing (i.e., SOS inflection point), while b determines the rate of seasonal transition, with greater absolute values of b indicating accelerated growth kinetics. Critically, the additive term $a + nb$ advances the curve by n days toward spring, whereas $a - nb$ delays it by n days toward summer under the constraint $b < 0$ and $n > 0$. Parameter c quantifies the seasonal amplitude of the fitted curve, and d represents the baseline offset, typically attributed to soil background interference in optical measurements. By definition, the sum $c + d$ equals the asymptotic maximum of the logistic curve.

2.2. PROSAIL and simulation of VIs

The PROSAIL model (Jacquemoud et al., 2009), a widely adopted physical modeling framework, has demonstrated robust performance in simulating canopy reflectance spectra and retrieving biophysical parameters for DBF across multiple studies (Sinha et al., 2020; Wan et al., 2024; Xu et al., 2022b; Zhang et al., 2006). This hybrid architecture couples the leaf-level PROSPECT-5 optical model (Feret et al., 2008) with the canopy-scale 4SAIL scattering model (Goel and Thompson, 2000), where 4SAIL utilizes PROSPECT-generated leaf reflectance/transmittance spectra as inputs to simulate hyperspectral canopy reflectance. PROSAIL uniquely incorporates both leaf-level biochemical traits (e.g., pigment concentrations) and canopy architectural parameters (e.g., leaf angle distribution), enabling simulations across a 400–2500 nm spectral range at 1 nm resolution (Jacquemoud and Baret, 1990).

The PROSAIL model parameters adopted in this study were collected from the published literatures of DBF studies, including leaf structure mesophyll (N, unitless), leaf carotenoid content (Car, $\mu g/cm^2$), leaf brown pigment content (C_{brown}, $\mu g/cm^2$), leaf equivalent water thickness (C_w,cm), leaf dry mass content (C_m, g/cm^2), leaf inclination determination a ($LIDF_a$, unitless), leaf inclination determination b ($LIDF_b$, unitless), ratio of leaf size to canopy height (hotspot, m/m), and soil brightness factor (B_s, unitless). To simplify the model inputs, some parameters were set to default values: the fraction of diffuse incoming solar radiation (skyl, unitless), which is derived by empirical formulas and has minimal impact on the simulated reflectance spectra (Jacquemoud, 1993); the relative azimuth angle (φ_{sv}, Degree), fixed following established DBF studies (Darvishzadeh et al., 2008); the sun zenith angle (θ_s, Degree), which is set to 30°; and the view zenith angle (θ_v,Degree), which is set to 10°. Details of the PROSAIL model's parameters and corresponding references are listed in Table 1.

Twenty-one VIs commonly employed in phenological monitoring were calculated from simulated spectra. To replicate satellite observation conditions, 20 VIs were convolved with MODIS's spectral response functions, supplemented by Red Edge bands and the narrowband photochemical reflectance index (PRI). Complete spectral bands, VI formulations and associated references are cataloged in Table 2.

2.3. Effects of leaf area and pigmental development to VI-based SOS

This study investigated whether VI-derived SOS is predominantly governed by canopy leaf area or pigmental phenology. We hypothesize that leaf area SOS (measured by LAI) precedes pigmental SOS (measured by LCC). Utilizing the PROSAIL model, we simulated time series for 21 VIs and extracted SOS using five widely used approaches, including threshold-based methods with thresholds of 10%, 20% and 50% (Chen et al., 2018; Shang et al., 2017), single logistic function with inflection point (Zhang et al., 2003), and direct maximum VI ratio algorithm (Cong et al., 2012).

To isolate leaf area and pigmental effects, we designed perturbation experiments:

1) Leaf area manipulation: LAI temporal trajectories were systematically advanced/delayed (1–10 day increments) while maintaining original LCC series.

2) Pigmental manipulation: LCC curves were temporally shifted under fixed LAI conditions.

SOS values were recalculated under each modified scenario (Eqs. 2–3):

$$E^T_{LAI(n)} = SOS^T_{(LAI(n),LCC)} - SOS^T_{(LAI,LCC)}\# \quad (2)$$

$$E^T_{LCC(n)} = SOS^T_{(LAI,LCC(n))} - SOS^T_{(LAI,LCC)}\# \quad (3)$$

where n denotes temporal offset (days), and superscript T indicates the 20% threshold. Positive/negative E_{LAI} values reflect SOS advancement/delay due to LAI shifts, with analogous interpretation for E_{LCC}.

The differential sensitivity metric δSOS was computed as:

$$\delta SOS = |E^T_{LAI(n)}| - |E^T_{LCC(n)}|\# \quad (4)$$

Positive δSOS values indicate VI-based SOS dominance by leaf area phenology, while negative values suggest pigmental control. Magnitude quantifies relative sensitivity, enabling strategic VI selection for specific phenological monitoring objectives.

2.4. Analysis strategies

To validate our hypotheses and assess the reliability of results, we conducted comparative analyses between numerical simulations and multi-source observations:

1) Field-derived SOS: At all three study sites, we fitted observed LAI and LCC time series with a single logistic function, and from these fitted

Table 1
Input parameters of the PROSAIL model.

Model	Variable	Description	Value	Unit	Reference
PROSPECT	N	Leaf mesophyll	1.5	unitless	(Xu et al., 2022a)
	LCC	Leaf chlorophyll content	Temporal measurements	$\mu g/cm^2$	(Gond et al., 1999; Brown et al., 2019; Yang et al., 2017)
	Car	Leaf carotenoids content	10	$\mu g/cm^2$	(Yang et al., 2021)
	Cbrown	Leaf brown pigments content	0	$\mu g/cm^2$	(Yang et al., 2021)
	Cw	Leaf equivalent water thickness	0.02	cm	(Yang et al., 2021)
	Cm	Leaf dry mass content	0.01	g/cm^2	(Yang et al., 2021)
SAIL	LIDFa	Leaf inclination determination a	−0.35	unitless	(Yang et al., 2021)
	LIDFb	Leaf inclination determination b	−0.15	unitless	(Yang et al., 2021)
	LAI	Leaf area index	Temporal measurements	m^2/m^2	(Gond et al., 1999; Brown et al., 2019; Yang et al., 2017)
	skyl	Fraction of diffuse incoming solar radiation	0.23	unitless	Default
	hotspot	Ratio of leaf size to canopy height	0.01	m/m	(Yang et al., 2021)
	Bs	Soil brightness factor	0.5	unitless	(Xu et al., 2022a)
	θ_s	Sun zenith angle	30	Degree	Default
	θ_v	View zenith angle	10	Degree	Default
	φ_{sv}	Relative azimuth angle	90	Degree	Default

Table 2
Overview of the vegetation indices (VIs) used for extracting SOS in this study.

No.	VIs	Description	Multispectral equations*	Reference
1	NDVI	Normalized Difference Vegetation Index	$\frac{NIR-R}{NIR+R}$	(Rouse et al., 1974)
2	EVI	Enhanced Vegetation Index	$2.5*\frac{NIR-R}{NIR+6*R-7.5*B+1}$	(Huete et al., 1997)
3	EVI2	Two-Band Enhanced Vegetation Index	$2.5*\frac{NIR-R}{NIR+2.4R+1}$	(Zhang et al., 2017)
4	GCC	Green Chromatic Coordinate	$\frac{G}{R+G+B}$	(Richardson, 2019)
5	SAVI	Soil-Adjusted Vegetation Index	$1.5*\frac{NIR-R}{NIR+R+0.5}$	(Huete, 1988)
6	CCI	Chlorophyll/Carotenoid Index	$\frac{G-R}{G+R}$	(Wong et al., 2019)
7	NIRv	Near-Infrared Reflectance of Vegetation	$\frac{NIR-R}{NIR+R}*NIR$	(Badgley et al., 2017)
8	NDII	Normalized Difference Infrared Index	$\frac{NIR-SWIR2}{NIR+SWIR2}$	(Hardisky et al., 1983)
9	NDPI	Normalized Difference Phenology Index	$\frac{NIR-(0.74*R+0.26*SWIR2)}{NIR+(0.74*R+0.26*SWIR2)}$	(Wang et al., 2017)
10	MSAVI	Modified SAVI	$0.5*\left[2*NIR+1-\sqrt{(2*NIR+1)^2-8*(NIR-R)}\right]$	(Qi et al., 1994)
11	PRI	Photochemical Reflectance Index	$\frac{\rho_{531}-\rho_{570}}{\rho_{531}+\rho_{570}}$	(Gamon et al., 1992)
12	MCARI	Modified Chlorophyll Absorption Ratio Index	$[(RE1-R)-0.2*(RE1-G)]*\left(\frac{RE1}{R}\right)$	(Daughtry et al., 2000)
13	CCII	Transformed Chlorophyll Absorption in Reflectance Index/Optimized Soil-Adjusted Vegetation Index	$\frac{3*[(RE1-R)-0.2*(RE1-G)*(RE1/R)]}{(1+0.16)*(NIR-R)/(NIR+R+0.16)}$	(Wu et al., 2008; Qian et al., 2022)
14	PI	Phenology Index	$\begin{cases} 0, if\ NDVI\ or\ NDII<0 \\ NDVI^2-NDII^2 \\ 0, if\ PI<0 \end{cases}$	(Gonsamo et al., 2012)
15	GNDVI	Green Normalized Difference Vegetation Index	$\frac{NIR-G}{NIR+G}$	(Gitelson et al., 1996)
16	SR	Simple Ratio	$\frac{NIR}{R}$	(Jordan, 1969)
17	NDRE	Normalized Difference Red-Edge Index	$\frac{NIR-RE1}{NIR+RE1}$	(Khodjaev et al., 2024)
18	MTCI	MERIS Terrestrial Chlorophyll Index	$\frac{RE2-RE1}{RE1-R}$	(Dash and Curran, 2004)
19	CVI	Chlorophyll Vegetation Index	$\frac{NIR*R}{G^2}$	(Vincini and Frazzi, 2011)
20	GCVI	Green Chlorophyll Vegetation Index	$\frac{NIR}{G}-1$	(Zhang et al., 2019)
21	CIre	Red Edge chlorophyll index	$\frac{NIR}{RE1}-1$	(Sun et al., 2018)

* Note: The B, G, R, NIR, SWIR2, RE1, and RE2 represent blue (459–479 nm), green (545–565 nm), red (620–670 nm), near infrared (841–876 nm), shortwave infrared (1628–1652 nm), red-edge 1 (698–713 nm), and red-edge 2 (733–748 nm) spectral bands, respectively.

curves, we further extracted leaf area SOS and pigmental SOS, corresponding to the onset of canopy leaf expansion and pigment dynamics, respectively.

2) Satellite-derived SOS: Guided by numerical simulation outputs, we identified two VIs demonstrating the strongest associations with leaf area and pigmental phenology, respectively. These VIs were computed from MODIS nadir BRDF-adjusted reflectance data (MCD43A4) over site-specific temporal windows: Harvard Forest (USA) & New Forest (UK): 2011–2020; De Inslag (Belgium): 2001–2010 (approximating the 1997 observational baseline). VIs time-series were smoothed using Singular Spectrum Analysis (SSA) (Keenan et al., 2014).
3) Broad-scale validation: To test generality beyond the focal sites, we examined VI-derived SOS differences across DBF, grassland, and cropland at mid-to-high latitudes in the Northern Hemisphere during 2011–2020. Five hundred random points per land cover type of DBF, grassland, and cropland were sampled from MODIS land cover maps (MCD12C1) (Fig. 2a and Supplementary Fig. S1a).

Both field and satellite SOS determinations employed the same five phenological extraction methods, ensuring methodological consistency. This multi-scale validation framework provided robust cross-examination of phenological drivers while accounting for differences between observational platforms.

3. Results

3.1. Comparison among the phenology extraction methods

Comparison among these methods shows that SOS estimates vary in absolute magnitude for each VI (Fig. 3a). The 10% threshold typically corresponds to the inflection point of seasonal green-up (~9.18%), while the 50% threshold occurs after the peak growth rate (Shang et al., 2017; Chen et al., 2018). Meanwhile, the 20% threshold generally produces SOS values close to the mean across all five methods. Despite differences in absolute timing, SOS values derived using the same method exhibit consistent trends across VIs (Fig. 3b), meaning that variation in SOS among indices is driven primarily by VI-specific characteristics rather than by the extraction method itself. This paper mainly presents the subsequent results using the 20%-threshold approach to ensure the results are concise and coherent. Results from the other methods and their comparisons are presented in the supplementary materials and discussion.

3.2. Indicative capabilities of VI-based SOS

Fig. 4 illustrates the differential sensitivity of VI-derived SOS to shifts in leaf area (LAI-driven) and pigmental (LCC-driven) phenology across the three sites. The simulations show that VI-based SOS exhibited proportional responses to temporal adjustments (1–10 day offsets) in LAI/

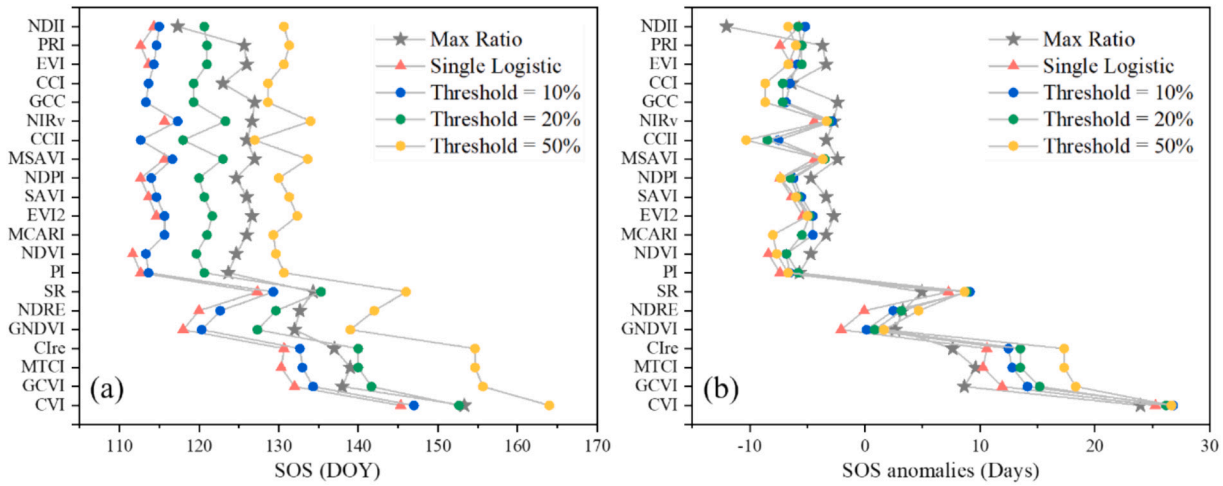


Fig. 3. VIs-derived SOS (a) using the diverse phenology extraction methods and (b) their anomalies for each of the methods across the three sites. The methods include a threshold-based approach with thresholds of 10% (blue points), 20% (green points), and 50% (yellow points), respectively; a single logistic method with the inflection point (orange triangles); and a direct maximum VI ratio algorithm (gray stars). The y-axis VIs are detailed in Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

LCC curves, with distinct divergence patterns between leaf area and pigmental indicators. Across all sites, most simulated VI-based SOS displayed pronounced sensitivity to LAI curve adjustments, indicating their strong capability in detecting leaf area SOS. In particular, NDII-based SOS displayed a near 1:1 correspondence with LAI shifts (i.e., a 10-day LAI shift produced ~9.8-day change in SOS), highlighting NDII's superior performance in tracking leaf area phenology. SOS derived from EVI, PRI, and CCII also demonstrated substantial indicative capability for leaf area expansion onset. In contrast, SOS estimates based on MTCI, CVI, GCVI, and CIre were more sensitive to LCC curve variations, reflecting a strong ability to capture the onset of leaf pigmental changes in DBF. Among these, CVI-based SOS showed the strongest sensitivity to LCC adjustments, indicating its particular effectiveness for pigmental phenology monitoring.

To systematically evaluate detection capabilities, Fig. 5 positions each VI according to its average sensitivity to leaf area versus pigmental SOS across all sites. The x/y-axis indicates the mean advanced days of VI-derived SOS in response to a 1–10-day advance in the original LAI/LCC series, relative to the unshifted baseline SOS. Higher values on the x-axis indicates greater sensitivity to leaf area expansion onset (LAI-driven), whereas higher values on the y-axis indicate greater sensitivity to pigmental onset (LCC-driven). The analysis reveals three distinct clusters: 1) Leaf aera-dominant VIs—14 out of the 21 VIs exhibited strong sensitivity to leaf area SOS (yellow zone), with NDII consistently outperforming all others; 2) Pigmental-dominant VIs—four VIs (MTCI, CVI, GCVI, and CIre) was primarily responsive to pigmental SOS, with CVI showing the highest sensitivity (green zone); 3) Dual but weak detectors—three VIs (SR, NDRE, and GNDVI) demonstrated comparable yet relatively low sensitivity to both leaf area and pigmental SOS (red zone).

3.3. Comparisons between field-observed and satellite-derived SOS

Fig. 6 present the observed LAI- and LCC-based SOS at three DBF sites, compared with NDII- and CVI-derived SOS obtained from MODIS land surface reflectance data. Two key findings emerge. First, leaf area phenology precedes pigmental development. LAI-based SOS (126 ± 8 days) occurred significantly earlier than LCC-based SOS (141 ± 11 days) ($p < 0.05$), directly supporting our hypothesis that canopy leaf expansion onset precedes pigmental change in DBFs. Second, strong agreement between field and satellite metrics. NDII-based SOS closely matched the onset of leaf area development, whereas CVI-based SOS aligned more strongly with pigmental initiation.

These field-validated patterns are consistent with simulation results presented in Section 3.1, confirming the capacity of VI-based metrics to distinguish leaf area and pigmental phenophases. MODIS-derived SOS further reinforced the structural-to-pigmental progression: NDII-based SOS (118 ± 14 days) preceded CVI-based SOS (149 ± 16 days) by an average of 31 days. While remote sensing estimates generally agreed with site-level observations, NDII-based SOS tended to occur slightly earlier than its LAI-based counterpart. Broad-scale analysis across mid-to-high latitude DBFs (Fig. 6d) yielded similar results. NDII-based SOS occurred earlier than CVI-based SOS at 86.1% of the DBF samples, with an average lead of approximately 20 days. This pattern extended to other land-cover types: in grasslands and croplands, NDII-based SOS preceded CVI-based SOS in 67.6% and 82.9% of samples, with average advances of approximately 14 and 19 days, respectively (Supplementary Fig. S1b-c).

4. Discussion

Satellite VIs are invaluable for monitoring land surface phenology across vast regions and time periods (Zeng et al., 2020; Ma et al., 2022). Yet a fundamental question remains unresolved: Do VI-derived SOS signals primarily reflect leaf area expansion or pigmental development? To address this, we integrated field-measured leaf area index (LAI) and leaf chlorophyll content (LCC) data with the PROSAIL model. We simulated 21 canopy-scale VIs under controlled scenarios where LAI and LCC variations were systematically decoupled. Our analysis reveals a clear separation between indices sensitive to leaf area versus pigmental development (Fig. 7). Specifically, 81% of VIs (e.g., NDII, EVI, and NIRv) detected SOS driven by leaf area development, while four VIs (CVI, MTCI, GCVI, and CIre) tracked SOS linked to pigmental development. Among these VIs, NDII emerged as the most robust indicator of leaf area expansion onset, while CVI best captured pigmental initiation—patterns corroborated by previous field observations and satellite data. These findings advance the ecological interpretation of SOS metrics in climate-impact studies.

4.1. How VI design shapes phenology detection

The division between leaf area- and pigmental-sensitive VIs is rooted in their fundamental design principles. Indices that heavily use near-infrared (NIR: 700–1300 nm) bands (e.g., NDVI, NDII) primarily capture structure aspects of plant phenology because canopy structures, including LAI, efficiently scatter NIR light (Wu et al., 2018). Structural

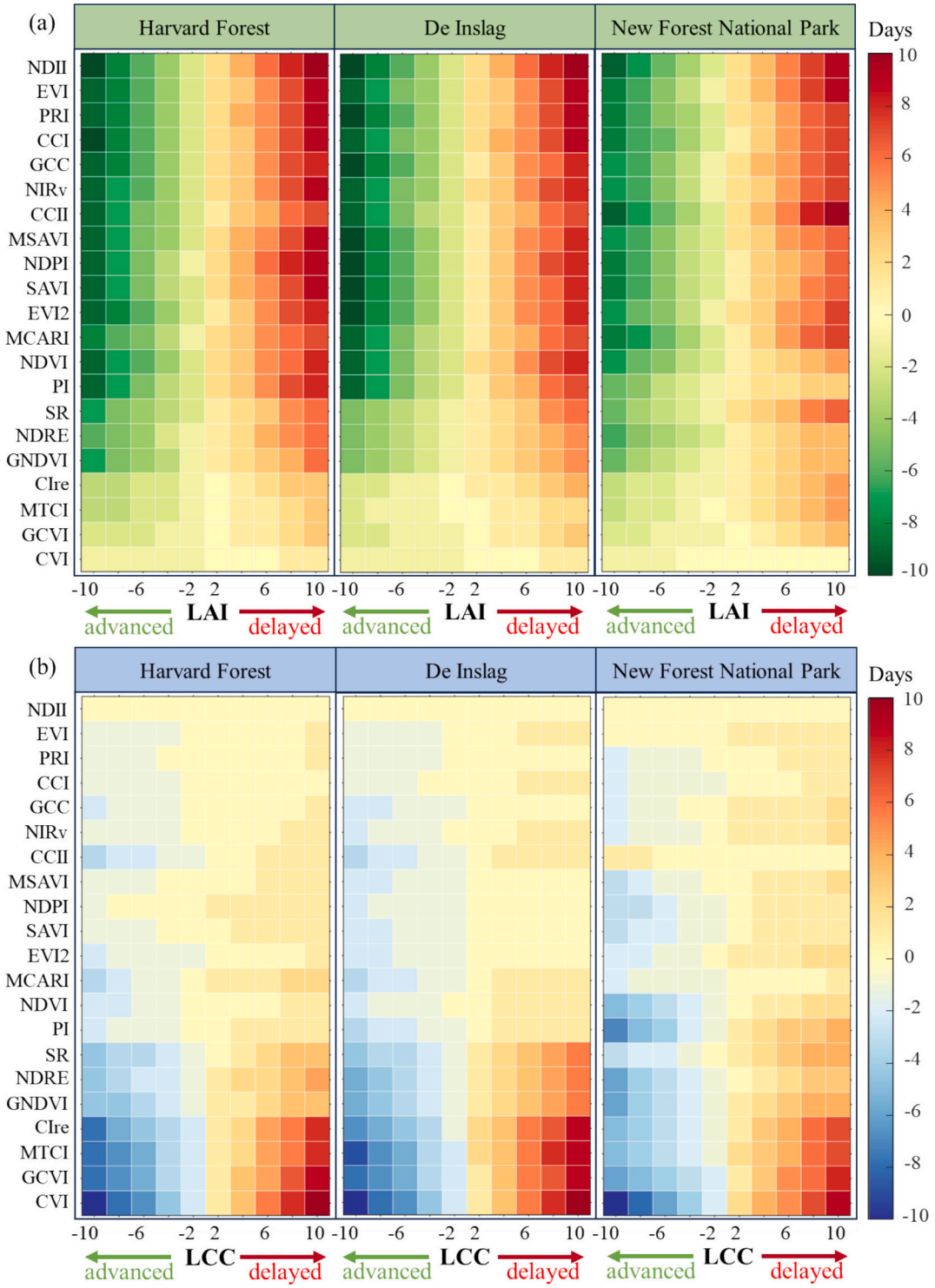


Fig. 4. Different Vegetation Index (VI)-based SOS in response to canopy leaf (a) area or (b) pigmental variabilities across the three sites. The SOS was derived from the VI time series using a threshold-based approach (20% threshold). The x-axis indicates the number of days by which the original (a) LAI or (b) LCC series were advanced (the negative values) or delayed (the positive values). The y-axis VIs are detailed in Table 2. The colors represent the difference between SOS derived from LAI/LCC changes and the initial SOS for each of the VIs. Larger absolute values indicate stronger influences of the leaf area (e.g., LAI) or pigmental trait (e.g., LCC).

complexity enhances NIR reflectance through multi-layer scattering, while chlorophyll absorption quickly saturates the red bands early in growth (Gao et al., 2023)—making NIR-heavy VIs particularly well suited for detecting structural SOS. Conversely, VIs using pigment-sensitive bands (400–700 nm) are designed to track biochemical trait changes. These spectral regions correlate with photosynthetic function, chlorophyll concentration, and light-use efficiency (Garbulsky et al., 2011; Qian et al., 2022). The MTCI index, for example, exploits the red-edge slope (705–753 nm) to track chlorophyll dynamics (Dash et al., 2010). However, there are exceptions, such as the PRI, which is formulated within a pigment-sensitive portion of the spectrum, yet PRI-derived SOS values show greater sensitivity to leaf area development.

NDII's strong leaf area expansion sensitivity, which was originally intended to detect leaf water content (1550–1750 nm), also known as

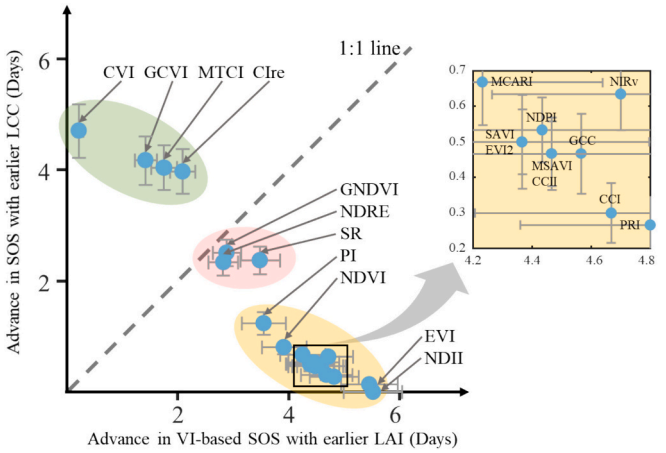


Fig. 5. Sensitivity of VI-derived SOS to shifts in LAI/LCC time series. The x/y-axis indicates the mean advanced days of SOS in response to a 1–10-day advance in the original LAI/LCC series, relative to the baseline (unshifted) SOS across the three study sites. Full names and definitions of the VIs are provided in Table 2.

the Land Surface Water Index (LSWI, Xiao et al., 2004), arises because canopy-level water absorption closely correlates with LAI that collectively define the higher-level reflectance spectra (Ceccato et al., 2001). Field validation using small unmanned aerial vehicle (sUAV) data confirms NDII's superior performance for detecting leaf emergence (Pan et al., 2024). In contrast, CVI's pigmental precision derives from its use of chlorophyll-sensitive green bands (530–570 nm), which minimize structural interference (Vincini et al., 2008). This narrow-band design allows CVI to isolate biochemical transitions more effectively than broad-band indices. Ultimately, the construction principles of a VI determine whether its SOS metric is more likely to capture the "leaf area" or "pigmental" aspect of spring phenology.

Notably, there is a trade-off: indices highly sensitive to leaf area phenology tend to perform poorly for pigmental phenology, and vice versa. Those with moderate sensitivity to both (e.g., SR, NDRE, and GNDVI) generally lack precision in detecting either process. This highlights that the challenge of relying on a single VI to accurately characterize both leaf area and pigment phenophases in deciduous broadleaf forests under varying environmental conditions.

4.2. How method selection shapes phenological interpretation

In this study, five methods were used to extract SOS: three threshold-based approaches, a single logistic method, and a maximum VI ratio algorithm. They have been shown to produce results highly consistent with other methods (Shen et al., 2024), For threshold-based approaches, a notable pattern emerges: the same VI can reflect fundamentally different biological processes depending on the threshold applied (Supplementary Fig. S2–S3). For example, GNDVI- and SR-based SOS consistently indicated leaf area expansion onset at low thresholds (10%–20%), but switched to pigmental initiation at the 50% threshold across all study sites. This threshold-dependent shift arises because VIs respond to changing plant priorities during growth (Pantin et al., 2012): 1) Early season (low thresholds)—resources are invested in structural construction at both leaf (e.g., leaf development and cell division) and canopy (e.g., leaf area expansion), where leaf area expansion dominates; 2) Mid-season (high thresholds): resources shifts toward biochemical

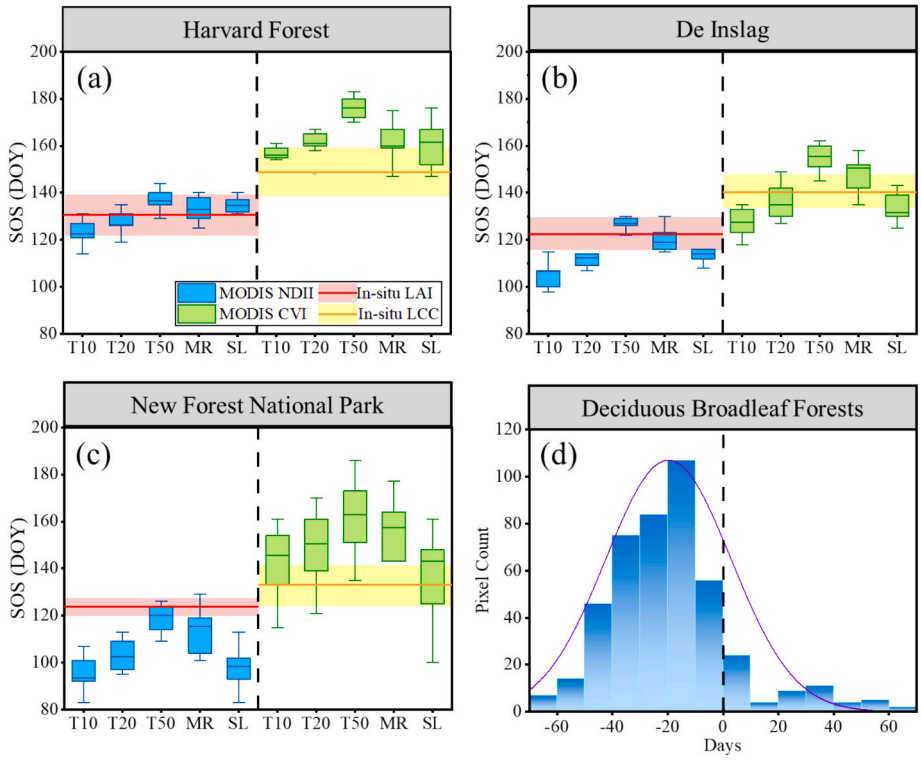


Fig. 6. Comparison of SOS derived from field observations (LAI and LCC) and satellite-based MODIS NDII/CVI using a threshold-based approach (20% threshold). (a) Three DBF study sites: Solid lines with shaded areas represent mean LAI- and LCC-based SOS (± standard error) obtained using five phenological extraction methods—T10, T20, T50 (threshold-based approaches with thresholds of 10%, 20%, and 50%, respectively), SL (single logistic), and MR (maximum VI ratio algorithm). Box plots show the distribution of VI-derived SOS from MODIS data: 2011–2021 for Harvard Forest, 2001–2011 for De Inslag, and 2011–2021 for New Forest National Park. Box plot elements follow standard definitions: lower whisker (Q1–1.5 IQR), lower quartile (Q1), median, upper quartile (Q3), and upper whisker (Q3 + 1.5 IQR), where IQR = Q3 − Q1. (b) Random samples ($n = 500$) from DBF at mid-to-high latitudes in the Northern Hemisphere, showing MODIS NDII- and CVI-derived SOS differences. (c) Frequency distribution of SOS differences between NDII and CVI for DBF samples.

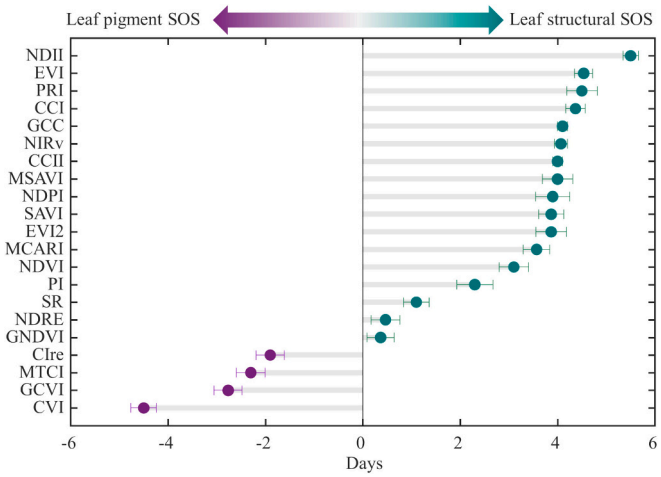


Fig. 7. Ranking of VIs by their ability to indicate structural versus pigmental SOS. Each point (± 1 SD error bars) shows the mean difference between VI-based SOS responses to variations in LAI and LCC across the three study sites, averaged over multiple phenological windows (see Fig. 3). Positive x-axis values indicate that the VI-based SOS is more closely aligned with structural SOS (driven by LAI changes), whereas negative values indicate stronger alignment with pigmental SOS (driven by LCC changes). The magnitude of the absolute value reflects the strength of the VI's indicative capability. VIs listed along the y-axis are defined in Table 2.

processes, particularly the enrichment of leaf pigment and other biochemical processes, amplifying pigmental influences on VIs. Consequently, 19 of 21 VIs (excluding NDII and CVI) showed increasing sensitivity to pigmental traits as thresholds rose from 10% to 50%. Nevertheless, NDII maintained leaf area-SOS fidelity across all thresholds, while CVI reliably tracked pigmental onset—highlighting their robustness for unambiguous phenology monitoring.

This threshold-induced variability underscores a key insight: VI-derived SOS is not a fixed biological event but a dynamic indicator whose ecological meaning depends on both index selection and extraction protocol. Researchers should therefore align threshold choices with specific study goals—low thresholds for leaf area development, higher thresholds for pigmental maturation—to ensure biologically consistent interpretations of land surface phenology.

4.3. Why scale transforms VI-based phenology interpretation

Unexpectedly, several chlorophyll-oriented indices such as PRI, MCARI, and CCII—originally designed to track leaf biochemistry—showed stronger links to leaf area phenology in our canopy-scale analysis. This apparent paradox highlights a critical but often overlooked fact: the phenological signals detected by VIs are inherently scale-dependent. While many VIs are developed based on leaf-level reflectance principles, satellite-based land surface phenology integrates the complexity of entire canopies, altering their ecological interpretation.

Take PRI as an example. At the leaf scale, PRI effectively tracks chlorophyll dynamics and photosynthetic activity through direct pigment-light interactions (Filella et al., 2009; Wong et al., 2019; Yang, 2024). Yet at the canopy scale, structural influences—such as leaf area, leaf angle distribution, and multi-layer scattering—can overwhelm PRI's pigmental sensitivity (Biriukova et al., 2020; Garbulsky et al., 2011; Wu et al., 2015). This explains why canopy PRI poorly represents leaf photosynthesis and aligns with our finding of weak PRI-SOS correlation. Moreover, while diurnal PRI time series can capture short-term physiology changes, over seasonal time frames (particularly during spring green-up), structural changes dominate its signal (Gitelson et al., 2017).

A similar scale effect occurs with MCARI. Although MCARI was designed to estimate chlorophyll content (Daughtry et al., 2000), it becomes heavily influenced by LAI at canopy scales (Wu et al., 2008). Even CCII, which is an improvement on MCARI that reduces soil background effects via OSAVI, remains structurally constrained when applied to satellite observations (Zhang et al., 2008). These examples reveal a consistent pattern: canopy structure can "drown out" biochemical signals, effectively repurposing many pigment-oriented VIs into leaf area indicators when scaled up.

This scale-driven transformation has practical implications for phenology research: 1) Leaf-optimized VIs (e.g., PRI, MCARI) may lose pigmental fidelity when applied to satellite imagery; 2) Universal VI interpretation across scales is unrealistic, as scale must be explicitly considered; 3) Even pigment-oriented indices like MCARI/CCII can remain value for canopy chlorophyll estimation but will remain influenced by structure during phenological transitions. Thus, when extracting phenology from satellite data, researchers should: 1) Prioritize VIs validated for canopy scales (e.g., NDII for leaf expansion, CVI for pigment dynamic); 2) Recognize the limitations of leaf-designed indices at larger scales; and 3) Interpret SOS metrics as scale-dependent signals rather than fixed biological events.

4.4. PROSAIL model and parametrization

Previous studies have shown that the PROSAIL model performs well in simulating VIs of DBFs (Xu et al., 2022b; Wan et al., 2024). In our analysis, phenological metrics derived from PROSAIL-simulated VIs closely match LAI- and LCC-based phenology at the three study sites and align well with metrics derived from site-specific satellite remote-sensing data. However, the PROSAIL model has inherent limitations compared with 3D radiative transfer models (RTMs), which may affect the robustness of the results.

To assess these limitations, we applied the LESS model, which is a 3D canopy RTM (Qi et al., 2019), at the Harvard Forest site to simulate canopy reflectances and VIs-based SOS, and compared the results with those from PROSAIL. The LESS simulations used forcing data and parameters derived from site-observed LAI and LCC, PROSAIL model parameters, and detailed tree attributes (e.g., coordinates, crown radius, DBH, height, and crown length) obtained from site records and Google Earth Engine. Five experimental scenarios were conducted: 1) original LAI and LCC series; 2–3) LAI series advanced by 5 and 10 days, respectively, with fixed LCC; and 4–5) LCC series advanced by 5 and 10 days, respectively, with fixed LAI.

Results showed that, relative to PROSAIL, LESS produced higher reflectamce in visible and shortwave bands but lower reflectance in near-infrared bands. This difference arises because, unlike the 1D PROSAIL model, LESS explicitly accounts for both LAI and canopy clumping, which increases transmittance, leading to an increase in the proportion of background (e.g., soil) reflectance relative to canopy reflectance under identical LAI (Li et al., 2025). Soil generally exhibits higher reflectance than vegetation in the visible and far-infrared bands, but lower reflectance in the near-infrared band. Despite these differences, canopy reflectances simulated by the two models were highly correlated ($r > 0.9$). Moreover, VIs and VIs-based SOS derived using both the relative threshold and inflection point methods were consistent between the two models, with correlation coefficient above 0.9, and exhibited similar sensitivities to the LAI- and LCC-based SOS (Fig. 8). These results suggest that, although PROSAIL does not fully represent the structural complexity of DBF canopies, its simulations remain robust and reliable for phenological application.

Notably, the PROSAIL model parameters used in this work were primarily obtained from literature values or default settings. To quantify their effects on VI-based SOS estimates, we conducted a systematic sensitivity analysis encompassing all model parameters. The results show that, relative to the effects of changes in LAI or LCC, the impact of individual parameters on phenological metric is relatively small, generally contributing less than 10% of variation induced by LAI or LCC

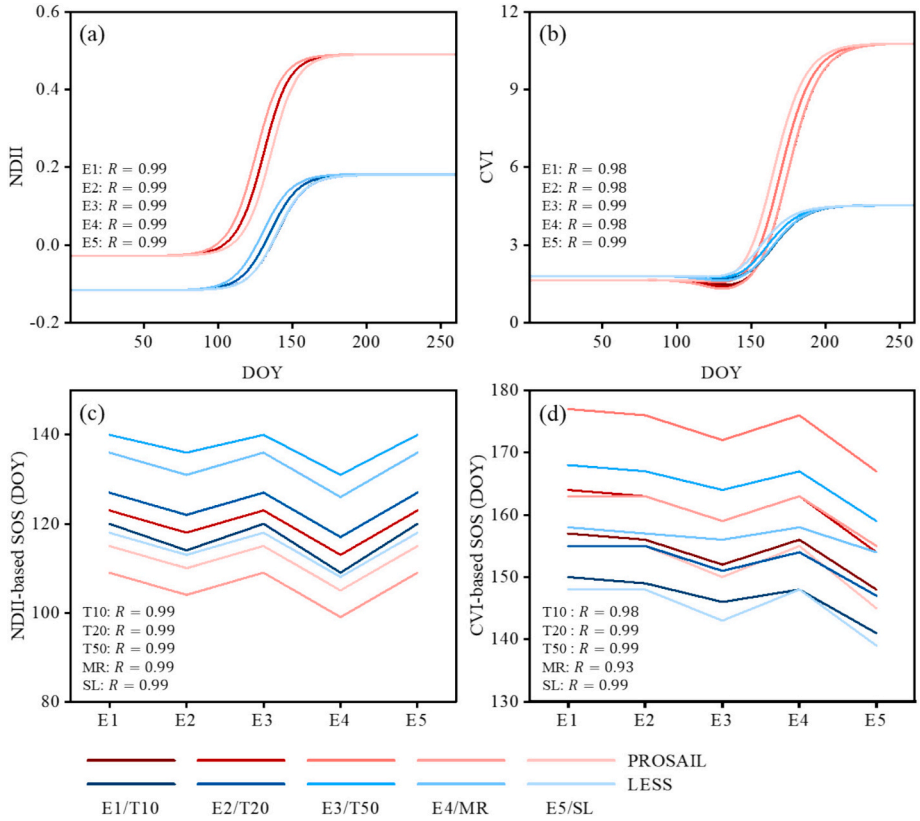


Fig. 8. Comparison of the (a-b) VIs (NDII and CVI) and (c-d) VIs-based SOS derived from the PROSAIL and LESS models. The colors represent the different experiments or phenology extraction methods. The five experimental scenarios are: (E1) the original LAI and LCC series; (E2–3) the LAI series advanced by 5 and 10 days, respectively, with fixed LCC; and (E4–5) the LCC series advanced by 5 and 10 days, respectively, with fixed LAI. The extraction methods include a threshold-based approach with thresholds of 10% (T10), 20% (T20), and 50% (T50), respectively; a direct maximum VI ratio algorithm (MR); and a single logistic method with an inflection point (SL).

changes (Supplementary Fig. S4-S6). This suggests that parameter uncertainty exerts only a limited influence on our main findings. Nevertheless, certain factors, such as soil properties (e.g., moisture and organic matter content) and illumination and view geometries, can affect simulated canopy reflectance. In particular, comparisons between the LESS and PROSAIL simulations show that when canopy clumping is explicitly represented, increased transmittance of radiation through the canopy can amplify the contribution of soil reflectance to the overall canopy spectrum. This effect can influence seasonal dynamics in vegetation indices (such as PRI, Yang, 2024) and, consequently, the derived phenological metrics. Therefore, incorporating site-specific, dynamically measured parameters remain essential for improving the accuracy and reliability of VI-based phenology studies.

4.5. *Bridging the interpretation gap in global phenology studies*

Our findings confirm that VIs encode distinct ecological processes, i.e., leaf area- and pigment-oriented phenology. This modality-specificity underscores the need for deliberate VI selection to ensure accurate biological interpretation. Yet a systematic bibliometric review of *Remote Sensing of Environment* publications from 2015 to 2024 reveals a critical disconnect: fully 64% of phenology studies (61/95 records) treat SOS as a generic "green-up" signal without specifying its biological meaning (Fig. 9). While these works effectively document large-scale phenological trends, such ambiguity obscures what SOS truly represents—introducing the risk of misattributing climate-driven changes to the wrong biological processes.

In contrast, the remaining 36% (34/95) explicitly link SOS to a clearly defined biological event, such as leaf emergence or photosynthesis onset, and validate their satellite-based metrics using field

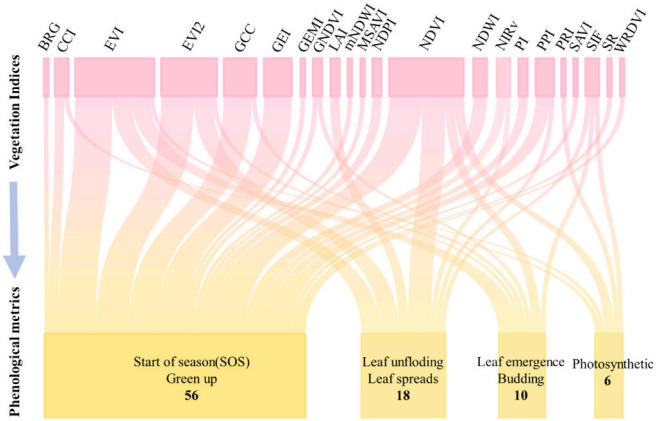


Fig. 9. Sankey diagram about the research of vegetation growing season start date, of which titles contain "Phenology", from 2015 to 2024 in the journal of *REMOTE SENSING OF ENVIRONMENT*. The details of the indices and phenological extraction methods are shown in Supplementary Table S2. The numbers in the phenological metrics denote amount of the records.

observations, near-surface cameras, or carbon flux measurements. For example, Zhao et al. (2022) used in situ observations to confirm that PlanetScope-derived EVI-based SOS captured leaf expansion in North American forests. Melaas et al. (2016) aligned Landsat EVI-based SOS to manual records of leaf emergence in temperate ecosystems. These targeted validations address part of the leaf area- vs. pigmental-phenology question, yet a systematic framework for distinguishing the two has

been lacking until now.

This study closes that gap by providing a robust evaluation of the ability of remotely sensed VI-based SOS to indicate both leaf area and pigmental phenology. Based on our findings, we recommend: 1) Using NDII to indicate leaf area expansion onset dates; 2) Using CVI to indicate pigmental onset dates; and 3) Explicitly defining the biological meaning of SOS in global phenology studies to avoid misinterpretation in climate impact assessments.

4.6. Uncertainties and further research

While our PROSAIL-based framework advances SOS interpretation, several key limitations warrant attention. First, model simulations were conducted exclusively for three Northern Hemisphere DBF sites. Broader validation using observational data from DBFs across a range of climates is needed to generalize these findings. Second, in this study LAI and LCC were manipulated independently and in a controlled, consistent manner, whereas in natural settings these variables often change simultaneously. Future work should therefore explore scenarios in which LAI and LCC vary simultaneously but over different timeframes or in opposite directions. Third, leaf area and pigment content represent only one facet of canopy structure and leaf physiology, respectively; additional structural and functional traits (e.g., leaf angle distribution, canopy clumping, water content, biochemical composition) can also influence canopy reflectance and thereby affect VI-based phenology. Last but not least, our conclusions are most applicable to deciduous vegetation with pronounced seasonal LAI changes. Evergreen ecosystems, where LAI varies little over the year, and highly heterogeneous canopies may exhibit different relationships between VI signals and phenological events, and thus warrant dedicated investigation. Future work should also extend analysis beyond SOS to other phenophases (e.g., peak greenness, end-of-season) to develop a comprehensive, universal framework for VI-based phenology interpretation.

5. Conclusions

Satellite-derived SOS metrics are often affected by spectral entanglement—the mixing of leaf area (measured by LAI) and pigmental (indicated by LCC) signals within reflectance data. To resolve this, we applied a PROSAIL simulation framework to deciduous broadleaf forests, testing 21 VIs. Our results show 2/3 of the VIs (14/21) primarily detect leaf area expansion onset, with NDII achieving near-perfect alignment. Conversely, four VIs (CVI, MTCI, GCVI, CIre) track pigmental onset, with CVI performing best. The remaining three VIs (SR, NDRE, GNDVI) provide mixed but relatively weak sensitivity to both leaf area expansion and pigmental onset. This divergence stems from: 1) Spectral design (NIR/SWIR bands for leaf area vs. red/red-edge for pigment), 2) Scale effects (canopy structure overrides leaf signals), and 3) Threshold choices (low thresholds favor leaf area; high thresholds emphasize pigment). Critically, 73% of recent studies (2015–2024, $n = 95$) label SOS vaguely as “green-up” without biological specificity. We thus urge that appropriate VIs for detecting land surface phenology should be adopted to distinguish leaf area and pigmental phenological metrics, to fulfill the specific research goals.

CRediT authorship contribution statement

Jiaxin Jin: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xiaolong Cheng:** Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation. **Yulong Cai:** Writing – review & editing, Visualization, Methodology. **Yuanwei Qin:** Writing – review & editing, Methodology, Funding acquisition. **Qiuan Zhu:** Writing – review & editing, Funding acquisition. **Weifeng Wang:** Writing – review & editing. **Peiqi Yang:** Writing – review & editing, Methodology. **Jianbo Qi:** Software, Methodology. **Feng Zhou:** Writing – review & editing. **Guishan Yang:** Writing – review & editing, Supervision. **Jin Wu:** Writing – review & editing, Visualization, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2026.115296.

Data availability

Data will be made available on request.

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